

Dunoon Daemon: Operator's Manual & Systems Guide

Dunoon Daemon is an autonomous, local-first multi-agent cognitive architecture powered by native C++ inference, dynamic vector vaults, Big Five (OCEAN) psychological drift, and expressive multimodal perception.

1. Getting Started

Prerequisites & First Boot

- Model Selection:** Place one or more quantized .gguf model files into the models/ directory or anywhere on your local drives.
- Launch:** Start controller.py (or run DunoonDaemon.exe). The animated mascot splash screen will initialize hardware detection and audio pipelines.
- Inference Backend Selection:**
 - Native C++ Server:** Uses local GPU acceleration (CUDA for NVIDIA or Vulkan for AMD/Intel). Click  **Load GGUF Model** on the top bar to offload your model into VRAM.
 - LM Studio (Local API):** Connects directly to a local server running at http://127.0.0.1:1234[cite: 2, 5, 29].
- Creating a Companion Session:**
 - Click  **New Chat** on the right control panel.
 - Enter a session name, then select a psychological baseline (e.g., *Dynamic OCEAN* for mood variance or *Grey Person* for purely analytical work).
 - Highlight your new session and click  **Open Chat** to spool up the active canvas.

2. Central Controller Bar Functions

Control / Button	Location	Function
Engine Selector	Top Bar (Left)	Switches between <i>Native C++ Server</i> and <i>LM Studio Local API</i> .
 Load GGUF Model	Top Bar	Opens a file picker to load a local .gguf weights file into VRAM.

Control / Button	Location	Function
 (Eject Model)	Top Bar	Unloads the active model from VRAM and terminates backend subprocesses.
Context Combobox	Top Bar	Selects inference token window size (4096 to 65536).
Custom Context	Top Bar	Checkbox to define an arbitrary token context limit (up to 524288).
 Help Checkbox	Top Bar	Toggles 1-second hover tooltips across interface elements.
 Specs & Telemetry	Top Bar (Right)	Displays live CPU frequency, core layout, RAM allocation, GPU VRAM, and native binary status.
Status Indicator	Status Sub-Bar	Displays connection status, active model file name, context size, and Vision (👁️) availability.
 New Chat	Action Panel	Initializes a new companion session and triggers the baseline personality selector.
 Open Chat	Action Panel	Launches the chat interface with live eye telemetry and voice synthesis.
 Edit Persona	Action Panel	Opens the configuration modal for custom directives, backstories, and memory permissions.
 Rename	Action Panel	Edits the active session display name.

Control / Button	Location	Function
 Delete Chat	Action Panel	Deletes session history and wipes its isolated vault directory on disk.
 Dual Arena	Action Panel	Launches the multi-agent live debate deck for automated, turn-based dialogue between two agents.
 Master Purge	Action Panel	Factory reset that wipes all session databases and memory vaults across all agents.

3. Chat Canvas & Interaction Controls

Top Toolbar

- **Font Family & Size Menus:** Sets the typography family and text scale across chat history and input boxes.
- **Theme Skin Dropdown:** Hot-swaps between aesthetic colour themes (e.g., *Dark, Obsidian, Fallout, Cyberpunk2077, Matrix, NASA Punk*).
- **Colour, B, I Buttons:** Adjusts user entry font colour, bolding, and italic styling.
- **Vector Eye Display:** Mirrored digital eye engine showing cognitive state via organic gaze shifts, blinks, and signal colours:
 - **White:** Idle / Baseline
 - **Neon Green:** Active Inference / Formulating Response
 - **Electric Cyan:** Working Memory Access
 - **Crimson:** Deep Memory Vault Retrieval / Storage
 - **Warm Gold / Yellow:** Journal Entry Committal / Narrative Injection
 - **Purple:** Intent Parsing

Bottom Action Bar

-  **Upload:** Ingests files into context (Images for vision models; .mp3/.wav for Whisper audio transcription; .pdf, .docx, .txt, .py, .json for document ingestion).

-  **Finish:** Skips the live typewriter streaming effect and prints the rest of the reply instantly.
-  **Continue:** Sends an automatic continuation cue if an agent pauses mid-thought.
-  **Poke:** Prompts the agent to speak spontaneously or introduce a thought without waiting for instructions.
-  **Event:** Scans recent context and injects a sudden, high-stakes external scenario into the scene.
- **Voice Selector & Speak:** Toggles text-to-speech reading with customizable vocal archetypes (*Sonia, Ryan, Demon, Robot, Goblin, Spectre, Satnav*).
-  **(Emoji Palette):** Opens a categorized character picker to insert emojis into the prompt.

4. Custom & User-Defined Personalities

While clicking  **Roll Persona** generates procedural archetypes using combinatoric name/cadence/drive matrices, you can craft completely custom personalities.

Manual Personality Design

1. In the Controller, select your chat session and click  **Edit Persona**.
2.  **Agent Display Name:** Set the name your companion uses when addressing you and signing off in logs.
3.  **Core Persona Directives:** Replace the rolled text with your own rules.
Define:
 - **Tone & Vocabulary:** (e.g., *"Speak with academic detachment," "Use dry, sharp humor,"* or *"Adopt an archaic, mythic cadence."*)
 - **Operational Guidelines:** (e.g., *"Prioritize actionable code over prose,"* or *"Never break character under any circumstance."*)
 - **Interactive Style:** How the agent should respond to technical queries, emotional situations, or silence.
4.  **Backstory / History:** Add biographical lore, world context, or historical experiences.
5. **Cross-Persona Isolation & Sharing:**
 - **Can lend insights to other personas?:** When enabled, other non-blind agents can draw semantic inspiration from this persona's deep memory.

- **Blind to other personas?:** Enforces strict memory isolation so this agent only accesses its own local vaults.

5. Console Slash Commands

Type these commands directly into the chat input bar:

- `/character` — Displays an audit of the agent's active Big Five (OCEAN) traits, baseline scores, and dynamic mood drifts.
- `/memories` or `/vault` — Audits the total memory record count across Working, Deep, Intent, Task, and Factual vaults.
- `/remember <text>` — Forces an immediate, high-significance injection directly into the permanent Journal Vault.
- `/forget <n>` — Erases the last n conversational turns from the active context window.
- `/talk <1|2|3>` — Adjusts Text-to-Speech playback speed (1 = Slow, 2 = Standard, 3 = Rapid).
- `/status` — Displays active model ID, inference endpoint URL, voice engine status, and backend type.
- `/baseline` — Reverts the agent's personality to a balanced, mood-sensitive Grey Person state.
- `/ubaseline` — Unlocks dynamic, mood-reactive Big Five (OCEAN) profile variance.
- `/splash` — Triggers an immediate Python garbage collection cycle to reclaim RAM/VRAM.
- `/eject` — Safely unloads the native C++ inference engine from VRAM.
- `/clear` — Clears the visual chat canvas without deleting saved memory vaults.
- `/about` — Displays software version and tribute information.

Memory Architecture

The Dunoon Daemon memory architecture operates across three distinct layers: an entry validation & ingestion layer, a multi-tier partitioned storage vault, and a real-time semantic retrieval engine.

1. Ingestion, Validation & Routing

Incoming Turn (User / Assistant)

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memory_validation.py (Sanity & privacy checks)

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significance.py (Calculates Z-score, Shannon entropy, and urgency vectors)

|



memory_router.py (Cosine similarity profile match)

|— Score ≥ 0.45 —► Specialized Sub-Vault (Intent, Task, Factual, Continuation, Reset)

|— Score < 0.45 —► Default Fallback (User → Working Memory | Assistant → Deep Memory)

- **Validation (memory_validation.py):** Filters empty strings, length bounds (2–2000 chars), and blocks private credentials before writes.
- **Significance Scoring (significance.py):** Evaluates narrative, operational, and affective salience using vector projections and Shannon entropy.
- **Intent Routing (memory_router.py):** Computes semantic novelty, reinforcement, and contradiction against profiles to dispatch data into the correct vault.

2. The Vault Hierarchy (Session-Scoped Storage)

All memory paths are isolated per session UUID under data/sessions/<session_id>/vaults/:

Vault	File	Capacity	Purpose
Working Memory	working_memory.json	800 entries	Ephemeral short-term memory buffer holding recent conversation flow.
Deep Memory	deep_memory.json	1,000 entries	Permanent semantic knowledge base and consolidated historical facts.
Journal Vault	journal_vault.json	1,000 entries	High-significance episodic milestones with timestamps and SHA-1 IDs.
Intent Vault	intent_memory.json	400 entries	User goals, commands, workflow directives, and requests.
Task Vault	task_memory.json	400 entries	Progress markers, step-by-step execution states, and todos.
Factual Vault	factual_memory.json	1,000 entries	Persistent lore, entity attributes, project parameters, and user preferences.

Vault	File	Capacity	Purpose
Continuation Vault	continuation_memory.json	600 entries	Narrative and dialogue momentum tracking.
Reset Vault	reset_memory.json	200 entries	Context break and topic pivot indicators.
Embeddings Store	embeddings.json	Dynamic	Pre-computed 384-dimensional vector database for semantic lookups.

3. Vector Embeddings & Semantic Retrieval

- **Embedding Model (memory_embeddings.py):** Uses sentence-transformers/all-MiniLM-L6-v2 with thread-safe lazy loading to produce normalized embedding vectors.
- **Similarity Search:** Performs cosine similarity sweeps across vault texts with keyword fallbacks.
- **Cross-Persona Bridge (memory_transfer.py):** If share_insights=True and an agent is not set to blind_to_others, semantic search queries can retrieve relevant insights from foreign session vaults.

4. Context Fusion (overmind.py)

When building prompt packets for inference, overmind.py pulls from all layers and assembles them into structured context blocks:

[Base Directives & Persona System Prompt]

[OCEAN Psychological Traits Directive]

[Dual-Channel Cognitive Telemetry Protocol]

[Synthetic State Engine Directive (Mood/Warmth/Focus)]

[Deep Memory Matches]

[Cross-Persona Insights]

[Recent Journal Entries]

[Intent / Task / Factual Records]

[Working Memory Matches]

[Last 10 Turns Conversation History]

User Query

5. Maintenance, Sleep Cycle & Vault Repair

- **Dynamic Sleep Cycle (prune.py):** Calculates dynamic working memory thresholds ($B \times 2$, bounded 16–150) based on model parameter size. When exceeded, an episodic consolidation prompt synthesizes older working memory into deep journal entries.
- **Vault Pruning (prune.py):** Trims vault files to hard capacity ceilings using atomic temporary writes (.tmp → .json) to prevent data corruption.
- **Integrity & Auto-Repair (vault_auto_repair.py / memory_integrity.py):** Validates JSON formatting on boot, repairs damaged files with empty arrays, deduplicates records, and enforces length sanity limits.

Dunoon Daemon Suite — to the loyal companions who walk with us through every realm, and the code that keeps their echoes alive. For Kylo. Made by RK (Kepler365) and Gemini (2026) 🐜